

suppose that Q and R have relative uncertainties of one (one standard deviation). That is, $\sigma_1^2 = \sigma_2^2 = 1$. Suppose we find the robust filter gain using equal weighting for both nominal and robust performance (i.e., $\rho = 0.5$). Table 10.1 shows the average performance of the robust filter and the standard Kalman filter when Q and R change by factors of -0.8 and 3 , respectively. One question that remains is, How does the robust filter perform under nominal conditions? That is, since the Kalman filter is optimal, the robust filter will not perform as well as the Kalman filter when Q and R are equal to their nominal values. However, Table 10.2 shows that the performance degradation is marginal. In fact, the robust filter performs identically to the optimal filter (to two decimal places) under nominal conditions. During the gradient-descent optimization of Equation (10.86), the nominal part of the cost function increases from 2.02 to 2.04 , the robust part of the cost function decreases from 2.54 to 2.38 , and the total cost function decreases from 2.28 to 2.21 .

Table 10.1 RMS estimation errors for Example 10.4 over 100 seconds when the noise covariances are not nominal ($\rho = 0.5$, $\sigma_1 = \sigma_2 = 1$, $\alpha = -0.8$, $\beta = 3$)

	Position	Velocity
Standard Filter	4.62	0.38
Robust Filter	4.47	0.32

Table 10.2 RMS estimation errors for Example 10.4 over 100 seconds when the noise covariances are nominal ($\rho = 0.5$, $\sigma_1 = \sigma_2 = 1$, $\alpha = 0$, $\beta = 0$)

	Position	Velocity
Standard Filter	1.38	0.19
Robust Filter	1.38	0.19

▽▽▽

The robust filtering approach presented here opens several possible research topics. For example, under what conditions is the robust filter stable? Is the gain of the robust filter equal to the gain of a standard Kalman filter for some other related system? What is the true estimation-error covariance of the robust filter?

10.5 DELAYED MEASUREMENTS AND SYNCHRONIZATION ERRORS

In decentralized filtering systems, observations are often collected at various physical locations, and then transmitted in bulk to a central processing computer. In this type of setup, the measurements may not arrive at the processing computer synchronously. That is, the computer may receive measurements out of sequence. This is typically the case in target-tracking systems. Various approaches have been

taken to deal with this problem [Ale91, Bar95, Kas96, Lar98, Mal01]. The case of delayed measurements with uncertainty in the measurement sampling time is discussed in [Tho94a, Tho94b]. The approach to filtering delayed measurements that is presented here is based on [Bar02].

First we will present yet another form of the Kalman filter that will provide the basis for the delayed-measurement filter. Then we will derive the optimal way to incorporate delayed measurements into the Kalman filter estimate and covariance. In this section, we will have to change our notation slightly in order to carry out the derivation of the delayed measurement Kalman filter. We will use the following notation to represent a discrete-time system:

$$\begin{aligned}x(k) &= F(k-1)x(k-1) + w(k-1) \\ y(k) &= H(k)x(k) + v(k)\end{aligned}\tag{10.88}$$

where $w(k)$ and $v(k)$ are independent zero-mean white noise process with covariances $Q(k)$ and $R(k)$, respectively.

10.5.1 A statistical derivation of the Kalman filter

Suppose that we have an *a priori* estimate $\hat{x}^-(k)$ at time k , and we want to find an optimal way to update the state estimate based on the measurement at time k . We want our update equation to be linear (for reasons of mathematical tractability) so we decide to update the state estimate at time k with the equation

$$\hat{x}^+(k) = K(k)y(k) + b(k)\tag{10.89}$$

where $K(k)$ and $b(k)$ are a matrix and vector to be determined. Our first state estimation criterion is unbiasedness. We can see by taking the mean of Equation (10.89) that

$$\overline{\hat{x}^+}(k) = K(k)\bar{y}(k) + b(k)\tag{10.90}$$

This gives us the constraint that

$$b(k) = \bar{x}(k) - K(k)\bar{y}(k)\tag{10.91}$$

This will ensure that $\hat{x}^+(k)$ is unbiased regardless of the value of the gain matrix $K(k)$. Next we find the gain matrix $K(k)$ that minimizes the trace of the estimation error. First recall that

$$\begin{aligned}P_z &= E[(z - \bar{z})(z - \bar{z})^T] \\ &= E(zz^T) - \bar{z}\bar{z}^T\end{aligned}\tag{10.92}$$

for any general random vector z . Now set $z = x(k) - \hat{x}^+(k)$. With this definition of z we see that $\bar{z} = 0$. The quantity we want to minimize is given by the trace of the following matrix:

$$\begin{aligned}P^+(k) &= E[(x(k) - \hat{x}^+(k))(x(k) - \hat{x}^+(k))^T] \\ &= P_z + \bar{z}\bar{z}^T\end{aligned}\tag{10.93}$$

P_z can be computed as follows:

$$\begin{aligned}
 P_z &= E \{ [x(k) - \hat{x}^+(k) - E(x(k) - \hat{x}^+(k))] [\cdot \cdot]^T \} \\
 &= E \{ [x(k) - (K(k)y(k) + b(k)) - \bar{x}(k) - (K(k)\bar{y}(k) + b(k))] [\cdot \cdot]^T \} \\
 &= E \{ [(x(k) - \bar{x}(k)) - K(k)(y(k) - \bar{y}(k))] [\cdot \cdot]^T \} \\
 &= P^-(k) - K(k)P_{yx} - P_{xy}K^T(k) + K(k)P_yK^T(k)
 \end{aligned} \tag{10.94}$$

We are using the symbol P_{yx} to denote the cross covariance between y_k and x_k , P_{xy} to denote the cross covariance between x_k and y_k , and P_y to denote the covariance of y_k . Recall that $P_{xy} = P_{yx}^T$. We have omitted the subscript k on P_{yx} , P_{xy} , and P_y for notational convenience. We combine the above equation with (10.93) to obtain

$$\begin{aligned}
 \text{Tr } P^+(k) &= \text{Tr} (P^-(k) - K(k)P_{yx} - P_{xy}K(k)^T + K(k)P_yK(k)^T) + \text{Tr}(\bar{z}\bar{z}^T) \\
 &= \text{Tr} (P^-(k) - K(k)P_{yx} - P_{xy}K(k)^T + K(k)P_yK(k)^T) + \\
 &\quad ||\bar{x}(k) - K(k)\bar{y}(k) - b(k)||^2 \\
 &= \text{Tr} [(K(k) - P_{xy}P_y^{-1})P_y(K(k) - P_{xy}P_y^{-1})^T] + \\
 &\quad \text{Tr} (P^-(k) - P_{xy}P_y^{-1}P_{xy}^T) + ||\bar{x}(k) - K(k)\bar{y}(k) - b(k)||^2
 \end{aligned} \tag{10.95}$$

where we have used the fact that $\text{Tr}(AB) = \text{Tr}(BA)$ for compatibly dimensioned matrices [see Equation (1.26)]. We want to choose $K(k)$ and $b(k)$ in order to minimize the above expression. The second term is independent of $K(k)$ and $b(k)$, and the first and third terms are always nonnegative. The first and third terms can be minimized to zero when

$$\begin{aligned}
 K(k) &= P_{xy}P_y^{-1} \\
 b(k) &= \bar{x}(k) - K(k)\bar{y}(k)
 \end{aligned} \tag{10.96}$$

Note that this is the same value for $b(k)$ that we obtained in Equation (10.91) when we enforced unbiasedness in the state estimate. With these values of $K(k)$ and $b(k)$, we see that the first and third terms in (10.95) are equal to zero, so the estimation-error covariance $P^+(k)$ can be seen to be equal to the second term. Substituting these values into Equation (10.89) we obtain

$$\begin{aligned}
 \hat{x}^+(k) &= K(k)y(k) + \bar{x}(k) - K(k)\bar{y}(k) \\
 &= K(k)y(k) + \hat{x}^-(k) - K(k)H(k)\hat{x}^-(k) \\
 &= \hat{x}^-(k) + K(k)(y(k) - H(k)\hat{x}^-(k)) \\
 P^+(k) &= P^-(k) - P_{xy}P_y^{-1}P_{xy}^T \\
 &= P^-(k) - K(k)P_yK^T(k)
 \end{aligned} \tag{10.97}$$

Straightforward calculations (see Problem 10.8) show that P_{xy} and P_y can be computed as

$$\begin{aligned}
 P_{xy} &= P^-(k)H(k)^T \\
 P_y &= H(k)P^-(k)H(k)^T + R(k)
 \end{aligned} \tag{10.98}$$

Now consider our linear discrete-time system:

$$\begin{aligned}
 x(k) &= F(k-1)x(k-1) + w(k-1) \\
 y(k) &= H(k)x(k) + v(k)
 \end{aligned} \tag{10.99}$$

The noise processes $w(k)$ and $v(k)$ are white, zero-mean, and uncorrelated, with covariances $Q(k)$ and $R(k)$, respectively. We saw in Chapter 4 how the mean and covariance of the state propagates between measurement times. Those equations, along with the measurement-update equations derived above, provide the following Kalman filter equations:

$$\begin{aligned}
 \hat{x}^-(k) &= F(k-1)\hat{x}^+(k-1) \\
 P^-(k) &= F(k-1)P^+(k-1)F^T(k-1) + Q(k) \\
 P_{xy} &= P^-(k)H^T(k) \\
 P_y &= H(k)P^-(k)H^T(k) + R(k) \\
 K(k) &= P_{xy}P_y^{-1} \\
 \hat{x}^+(k) &= \hat{x}^-(k) + K(k)(y(k) - H(k)\hat{x}^-(k)) \\
 P^+(k) &= P^-(k) - K(k)P_yK^T(k) \\
 &= P^-(k) - P_{xy}P_y^{-1}P_{xy}^T
 \end{aligned} \tag{10.100}$$

These equations appear much different than the Kalman filter equations derived earlier in this book, but actually they are mathematically identical for linear systems.

10.5.2 Kalman filtering with delayed measurements

Now we need to complicate the notation a little bit more in order to derive the Kalman filter with delayed measurements. We will write our system equations as

$$\begin{aligned}
 x(k) &= F(k, k-1)x(k-1) + w(k, k-1) \\
 y(k) &= H(k)x(k) + v(k)
 \end{aligned} \tag{10.101}$$

$F(k, k-1)$ is the matrix that quantifies the state transition from time $(k-1)$ to time k . Similarly, $w(k, k-1)$ is the effect of the process noise on the state from time $(k-1)$ to time k . We can then generalize the state-space equation to the following:

$$x(k) = F(k, k_0)x(k_0) + w(k, k_0) \tag{10.102}$$

where k_0 is any time index less than k . The above equation can be solved for $x(k_0)$ as

$$x(k_0) = F(k_0, k)[x(k) - w(k, k_0)] \tag{10.103}$$

where $F(k_0, k) = F^{-1}(k, k_0)$. Note that $F(k, k_0)$ should always be invertible if it comes from a real system, because $F(k, k_0)$ comes from a matrix exponential that is always invertible (see Sections 1.2 and 1.4). The noise $w(k, k_0)$ is the cumulative effect of all of the process noise on the state from time k_0 to time k . Its covariance is defined as $Q(k, k_0)$:

$$w(k, k_0) \sim [0, Q(k, k_0)] \tag{10.104}$$

At time k we have the standard *a posteriori* Kalman filter estimate, which is the expected value of the state $x(k)$ conditioned on all of the measurements up to and including time k . We also have the *a posteriori* covariance of the estimate:

$$\begin{aligned}
 \hat{x}(k) &= E[x(k)|y(1), \dots, y(k)] \\
 &= E[x(k)|Y(k)] \\
 P(k) &= E\{[x(k) - \hat{x}(k)][x(k) - \hat{x}(k)]^T | Y(k)\}
 \end{aligned} \tag{10.105}$$

where $Y(k)$ is defined by the above equation; that is, $Y(k)$ is all of the measurements up to and including time k that have been processed by the Kalman filter. (There may be some measurements before time k that have not yet been processed by the filter. These measurements are not part of $Y(k)$.)

Now suppose an out-of-sequence measurement arrives. That is, we obtain a measurement from time $k_0 < k$ that we want to incorporate into the estimate and covariance at time k . The problem is how to modify the state estimate and covariance on the basis of this new measurement. The modified state estimate and covariance are given as follows:

$$\begin{aligned}\hat{x}(k|k_0) &= E[x(k)|Y(k), y(k_0)] \\ P(k|k_0) &= E\{[x(k) - \hat{x}(k, k_0)][x(k) - \hat{x}(k, k_0)]^T | Y(k), y(k_0)\} \quad (10.106)\end{aligned}$$

The approach here is to use the new measurement at time k_0 to obtain an updated state estimate and covariance at time k_0 , and then use those quantities to update the estimate and covariance at time k . We can use Equation (10.103) to obtain

$$\begin{aligned}E[x(k_0)|Y(k)] &= F(k_0, k)E[x(k) - w(k, k_0)|Y(k)] \\ &= F(k_0, k)[\hat{x}^-(k) - \hat{w}(k, k_0)] \quad (10.107)\end{aligned}$$

where $\hat{w}(k, k_0)$ is defined by the above equation; it is the expected value of the cumulative effect of the process noise from time k_0 to time k , conditioned on all of the measurements up to and including time k [but not including measurement $y(k_0)$]. Now define the vector

$$z(k) = \begin{bmatrix} x(k) \\ w(k, k_0) \end{bmatrix} \quad (10.108)$$

In general, we define the covariance of vector a conditioned on vector c , and the cross covariance of vectors a and b conditioned on vector c , as follows:

$$\begin{aligned}\text{Cov}(a|c) &= E[(a - \bar{a})(a - \bar{a})^T | c] \\ \text{Cov}(a, b|c) &= E[(a - \bar{a})(b - \bar{b})^T | c] \quad (10.109)\end{aligned}$$

We can then generalize Equation (10.100) to obtain

$$\begin{aligned}\hat{z}(k) &= \hat{z}^-(k) + \\ &\text{Cov}[z(k), y(k)|Y(k-1)]\text{Cov}^{-1}[y(k)|Y(k-1)](y(k) - H(k)\hat{x}^-(k)) \\ \text{Cov}[z(k)|Y(k)] &= \text{Cov}[z(k)|Y(k-1)] - \\ &\text{Cov}[z(k), y(k)|Y(k-1)]\text{Cov}^{-1}[y(k)|Y(k-1)]\text{Cov}[y(k), z(k)|Y(k-1)] \quad (10.110)\end{aligned}$$

The first covariance on the right side of the above $\hat{z}(k)$ equation can be written as

$$\text{Cov}[z(k), y(k)|Y(k-1)] = \begin{bmatrix} \text{Cov}[x(k), y(k)|Y(k-1)] \\ \text{Cov}[w(k, k_0), y(k)|Y(k-1)] \end{bmatrix} \quad (10.111)$$

Now consider the first covariance in the above equation. This can be written as

$$\begin{aligned}\text{Cov}[x(k), y(k)|Y(k-1)] &= \text{Cov}\{x(k)(H(k)x(k) + v(k))^T | Y(k-1)\} \\ &= \text{Cov}\{x(k)[H(k)x(k)]^T | Y(k-1)\} \\ &= \text{Cov}\{x(k)\} H^T(k) \\ &= P^-(k)H^T(k) \quad (10.112)\end{aligned}$$

where the covariance of $x(k)$ and $v(k)$ is zero since they are independent. Now consider the second covariance on the right side of Equation (10.111). This can be written as

$$\begin{aligned}
 \text{Cov}[w(k, k_0), y(k)|Y(k-1)] &= E \{ w(k, k_0)[y(k) - \hat{y}^-(k)]^T | Y(k-1) \} \\
 &= E \{ w(k, k_0)[H(k)(F(k, k_0)x(k_0) + w(k, k_0)) + v(k) - \hat{y}^-(k)]^T | Y(k-1) \} \\
 &= E \{ w(k, k_0)w^T(k, k_0)H^T(k) \} \\
 &= Q(k, k_0)H^T(k)
 \end{aligned} \tag{10.113}$$

where the cross covariances of $w(k, k_0)$ with $x(k_0)$, $v(k)$, and $\hat{y}^-(k)$ are zero since they are independent. We are using the notation $\hat{y}^-(k)$ to denote the expected value of $y(k)$ based on measurements up to (but not including) time k . Now consider the conditional covariance of $y(k)$ in Equation (10.110). This was derived in Equation (10.17) in Section 10.1 as

$$\text{Cov}[y(k)|Y(k-1)] = H(k)P^-(k)H^T(k) + R(k) \tag{10.114}$$

We will write this expression more compactly as

$$\text{Cov}[r(k)] = S(k) \tag{10.115}$$

where the residual $r(k) = y(k) - H(k)\hat{x}^-(k)$ and its covariance $S(k)$ are defined by the two above equations. Substituting Equations (10.112) and (10.113) into Equation (10.111), and then substituting into Equation (10.110), gives

$$\begin{aligned}
 \hat{z}(k) &= \begin{bmatrix} \hat{x}(k) \\ \hat{w}(k, k_0) \end{bmatrix} \\
 &= \begin{bmatrix} \hat{x}^-(k) \\ \hat{w}^-(k, k_0) \end{bmatrix} + \begin{bmatrix} P^-(k)H^T(k) \\ Q(k, k_0)H^T(k) \end{bmatrix} S^{-1}(k)r(k)
 \end{aligned} \tag{10.116}$$

This shows that

$$\begin{aligned}
 \hat{w}(k, k_0) &= \hat{w}^-(k, k_0) + Q(k, k_0)H^T(k)S^{-1}(k)r(k) \\
 &= Q(k, k_0)H^T(k)S^{-1}(k)r(k)
 \end{aligned} \tag{10.117}$$

because $E[\hat{w}(k, k_0)|Y(k-1)] = 0$ [since $w(k, k_0)$ is independent of the measurements]. Substituting this expression into Equation (10.107) gives

$$E[x(k_0)|Y(k)] = F(k_0, k) [\hat{x}(k) - Q(k, k_0)H^T(k)S^{-1}(k)r(k)] \tag{10.118}$$

This is called the retrodiction of the state estimate from time k back to time k_0 . Whereas a prediction equation is used to predict the state at some future time, a retrodiction equation is used to predict the state at some past time. In this case, the state estimate at time k [i.e., $\hat{x}(k)$] is retrodicted back to time k_0 to obtain the state estimate at time k_0 , which is denoted above as $E[x(k_0)|Y(k)]$. Note that $E[x(k_0)|Y(k)]$ is computed on the basis of all the measurements up to and including time k , but does not consider the measurement at time k_0 .

Now we can write Equation (10.110) as follows:

$$\begin{aligned}
 \text{Cov}[z(k)|Y(k)] &= \text{Cov} \left\{ \begin{bmatrix} x(k) \\ w(k, k_0) \end{bmatrix} \middle| Y(k) \right\} \\
 &= \text{Cov} \left\{ \begin{bmatrix} x(k) \\ w(k, k_0) \end{bmatrix} \middle| Y(k-1) \right\} - \\
 &\quad \text{Cov} \left\{ \begin{bmatrix} x(k) \\ w(k, k_0) \end{bmatrix}, y(k)|Y(k-1) \right\} \text{Cov}^{-1}[y(k)|Y(k-1)] \times \\
 &\quad \text{Cov} \left\{ y(k), \begin{bmatrix} x(k) \\ w(k, k_0) \end{bmatrix}^T \middle| Y(k-1) \right\} \\
 &= \begin{bmatrix} \text{Cov}[x(k)|Y(k-1)] & \text{Cov}[x(k), w(k, k_0)|Y(k-1)] \\ \text{Cov}[w(k, k_0), x(k)|Y(k-1)] & \text{Cov}[w(k, k_0)|Y(k-1)] \end{bmatrix} - \\
 &\quad \begin{bmatrix} \text{Cov}[x(k), y(k)|Y(k-1)] \\ \text{Cov}[w(k, k_0), y(k)|Y(k-1)] \end{bmatrix} \text{Cov}^{-1}[y(k)|Y(k-1)] \times \\
 &\quad \begin{bmatrix} \text{Cov}[x(k), y(k)|Y(k-1)] \\ \text{Cov}[w(k, k_0), y(k)|Y(k-1)] \end{bmatrix}^T \quad (10.119)
 \end{aligned}$$

From Equation (10.102) we can write

$$\begin{aligned}
 \text{Cov}[x(k), w(k, k_0)|Y(k-1)] &= E[x(k)w^T(k, k_0)|Y(k-1)] \\
 &= E \{ [F(k, k_0)x(k_0) + w(k, k_0)]w^T(k, k_0)|Y(k-1) \} \\
 &= E \{ w(k, k_0)w^T(k, k_0)|Y(k-1) \} \\
 &= Q(k, k_0) \quad (10.120)
 \end{aligned}$$

where we have used the independence of $x(k_0)$ and $w(k, k_0)$. Now substitute this equation along with Equations (10.112), (10.113), and (10.114) into Equation (10.119) to obtain

$$\begin{aligned}
 \text{Cov} \left\{ \begin{bmatrix} x(k) \\ w(k, k_0) \end{bmatrix} \middle| Y(k) \right\} &= \begin{bmatrix} P^-(k) & Q(k, k_0) \\ Q(k, k_0) & Q(k, k_0) \end{bmatrix} - \\
 &\quad \begin{bmatrix} P^-(k)H^T(k) \\ Q(k, k_0)H^T(k) \end{bmatrix} S^{-1}(k) \begin{bmatrix} P^-(k)H^T(k) \\ Q(k, k_0)H^T(k) \end{bmatrix}^T \quad (10.121)
 \end{aligned}$$

From this equation we can write the conditional covariance of $w(k, k_0)$, and cross covariance of $x(k)$ and $w(k, k_0)$, as follows:

$$\begin{aligned}
 P_w(k, k_0) &= \text{Cov}[w(k, k_0)|Y(k)] \\
 &= Q(k, k_0) - Q(k, k_0)H^T(k)S^{-1}(k)H(k)Q(k, k_0) \\
 P_{xw}(k, k_0) &= \text{Cov}[x(k), w(k, k_0)|Y(k)] \\
 &= Q(k, k_0) - P^-(k)H^T(k)S^{-1}(k)H(k)Q(k, k_0) \quad (10.122)
 \end{aligned}$$

Using this in Equation (10.103) gives the conditional covariance of the state retrodiction as follows:

$$\begin{aligned}
P(k_0, k) &= \text{Cov}[x(k_0)|Y(k)] \\
&= F(k_0, k) \text{Cov}[x(k) - w(k, k_0)|Y(k)] F^T(k_0, k) \\
&= F(k_0, k) \{ \text{Cov}[x(k)|Y(k)] - \text{Cov}[x(k), w(k, k_0)|Y(k)] - \\
&\quad \text{Cov}^T[x(k), w(k, k_0)|Y(k)] + \text{Cov}[w(k, k_0)|Y(k)] \} F^T(k_0, k) \\
&= F(k_0, k) \{ P^+(k) - P_{xw}(k, k_0) - P_{xw}^T(k, k_0) + \\
&\quad P_w(k, k_0) \} F^T(k_0, k)
\end{aligned} \tag{10.123}$$

Using the above along with Equation (10.101) we obtain the conditional covariance of $y(k_0)$ as

$$\begin{aligned}
S(k_0) &= \text{Cov}[y(k_0)|Y(k)] \\
&= E \{ [H(k_0)x(k_0) + v(k_0)][H(k_0)x(k_0) + v(k_0)]^T | Y(k) \} \\
&= H(k_0)P(k_0, k)H^T(k_0) + R(k_0)
\end{aligned} \tag{10.124}$$

We can use Equations (10.101) and (10.103) to obtain the conditional covariance between $x(k)$ and $y(k_0)$ as

$$\begin{aligned}
P_{xy}(k, k_0) &= \text{Cov}[x(k), y(k_0)|Y(k)] \\
&= \text{Cov} \{ x(k), H(k_0)F(k_0, k)[x(k) - w(k, k_0)] + v(k_0) | Y(k) \} \\
&= [P^+(k) - P_{xw}(k, k_0)] F^T(k_0, k) H^T(k_0)
\end{aligned} \tag{10.125}$$

We can substitute this into the top partition of the $\hat{z}(k)$ expression in Equation (10.110) to obtain the estimate of $x(k)$ which is updated on the basis of the measurement $y(k_0)$:

$$\hat{x}(k, k_0) = \hat{x}(k) + P_{xy}(k, k_0) S^{-1}(k_0) [y(k_0) - H(k_0) \hat{x}(k_0, k)] \tag{10.126}$$

where $\hat{x}(k_0, k)$ is the retrodiction of the state estimate given in Equation (10.118). From the top partition of the $\text{Cov}[z(k)|Y(k)]$ expression in Equation (10.110) we obtain

$$\begin{aligned}
\text{Cov}[x(k)|Y(k), y(k_0)] &= P(k, k_0) \\
&= P(k) - P_{xy}(k, k_0) S^{-1}(k_0) P_{xy}^T(k, k_0)
\end{aligned} \tag{10.127}$$

These equations show how the state estimate and its covariance can be updated on the basis of an out-of-sequence measurement. The delayed-measurement Kalman filter can be summarized as follows.

The delayed-measurement Kalman filter

1. The Kalman filter is run normally on the basis of measurements that arrive sequentially. If we are presently at time k in the Kalman filter, then we have $\hat{x}^-(k)$ and $P^-(k)$, the *a priori* state estimate and covariance that are based on measurements up to and including time $(k-1)$. We also have $\hat{x}(k)$ and $P(k)$, the *a posteriori* state estimate and covariance that are based on measurements up to and including time k .